

Jenkins 4 task 1

1. Setup Jenkins CI/CD pipeline using freestyle job using Docker containers using below code. <https://github.com/betawins/hiring-app.git> Stages:

- > Git Clone
- > SonarQube Integration
- > Maven Compilation
- > Nexus Artifactory
- > Slack Notification
- > Deploy On tomcat

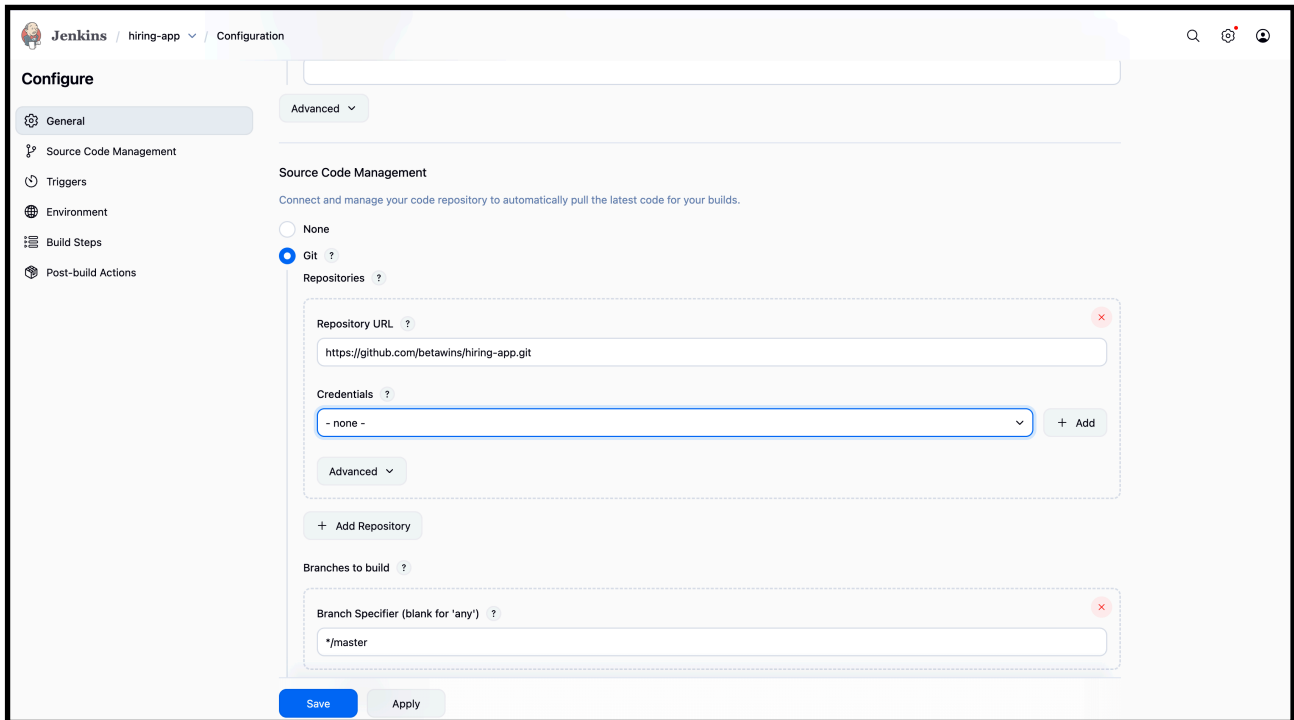
A). Git clone

Git clone to download the source code.

Follow the below steps

- > step 1 :
- > open your Jenkins dashboard
 - Go to your Jenkins URL → `h3p://<your-server-ip>:8080/`
- > **iam using 8081 as port number for my Jenkins application**
- > create a job —> inside general —> source code manager —> attach repo

1. <https://github.com/betawins/hiring-app.git>



—-> click save

—-> now go to the Jenkins workspace and check the files available from the repo

```
[root@ip-172-31-65-24 jenkins]# cd workspace
[root@ip-172-31-65-24 workspace]# ls
first-job hiring-app job2 job3 private private@tmp vprofile
[root@ip-172-31-65-24 workspace]# cd hiring-app
[root@ip-172-31-65-24 hiring-app]# ll
total 0
[root@ip-172-31-65-24 hiring-app]# ll
total 24
-rw-r--r--. 1 jenkins jenkins 476 Dec 29 11:59 Dockerfile
-rw-r--r--. 1 jenkins jenkins 1757 Dec 29 11:59 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 88 Dec 29 11:59 README.md
-rw-r--r--. 1 jenkins jenkins 408 Dec 29 11:59 'Untitled Diagram.drawio'
-rw-r--r--. 1 jenkins jenkins 1801 Dec 29 11:59 jenkinsfile-cicd
-rw-r--r--. 1 jenkins jenkins 890 Dec 29 11:59 pom.xml
drwxr-xr-x. 3 jenkins jenkins 18 Dec 29 11:59 src
[root@ip-172-31-65-24 hiring-app]#
```

b) SonarQube Integration

Sonarqube Integration to check the quality of code

Step 1 :

- > launch a instance and
- > install java

```
Warning: Permanently added 'ec2-98-93-252-172.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
#_
~\ #####_ Amazon Linux 2023
~\ #####\
~\ #####|
~\ \###|
~\ \#/ https://aws.amazon.com/linux/amazon-linux-2023
~\ V~' '->
~\ /
~\ /
~\ /m/'

[[ec2-user@ip-172-31-95-0 ~]$ sudo -i
[[root@ip-172-31-95-0 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[[root@ip-172-31-95-0 ~]# postgresql115-server postgresql115 -y
-bash: postgresql115-server: command not found
[[root@ip-172-31-95-0 ~]# sudo dnf install -y java-17-amazon-corretto
Last metadata expiration check: 0:08:59 ago on Mon Dec 29 12:16:33 2025.
Dependencies resolved.

=====
Package                                     Architecture
=====
Installing:
  java-17-amazon-corretto                   x86_64
Installing dependencies:
  alsa-lib                                   x86_64
  cairo                                      x86_64
  dejavu-sans-fonts                          noarch
  dejavu-sans-mono-fonts                     noarch
```

systemctl start docker

docker pull sonarqube

docker run -d --name sonarqube -p 9000:9000 sonarqube:lts

Docker ps

```
[ec2-user@ip-172-31-89-80 ~]$ sudo -i
[root@ip-172-31-89-80 ~]# yum install docker -y
Last metadata expiration check: 0:00:19 ago on Mon Dec 29 07:55:25 2025.
Dependencies resolved.
=====
Package                                                    Architecture
=====
Installing:
  docker                                                    x86_64
Installing dependencies:
  container-selinux                                         noarch
  containerd                                                x86_64
  iptables-libs                                             x86_64
  iptables-nft                                              x86_64
  libcgroup                                                 x86_64
  libnetfilter_conntrack                                   x86_64
  libnfnetlink                                              x86_64
  libnftnl                                                  x86_64
  pigz                                                      x86_64
  runc                                                      x86_64
Transaction Summary
=====
Install 11 Packages

Total download size: 74 M
Installed size: 280 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm
```

```
Complete!
[root@ip-172-31-89-80 ~]# systemctl start docker
[root@ip-172-31-89-80 ~]# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
20043066d3d5: Pull complete
f878172f1ac3: Pull complete
0b92fa77023f: Pull complete
1321ce5b3e3a: Pull complete
4a0e444f3c26: Pull complete
f60cf9719a3e: Downloading [=====>] 436.6MB/901.5MB
2a0225763c5a: Download complete
4f4fb700ef54: Download complete
```



```
[root@ip-172-31-89-80 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
sonarqube     latest   247ce055a80b   2 weeks ago   1.27GB
[root@ip-172-31-89-80 ~]#
```

```
sonarqube     latest   247ce055a80b   2 weeks ago   1.27GB
[root@ip-172-31-89-80 ~]# docker run -itd -p 9000:9000 sonarqube
a164990fb54f07b09e6691d1d7bdaa9a8aad789af5641325ba31d43f3ba84b6b
[root@ip-172-31-89-80 ~]#
```

```
a164990fb54f07b09e6691d1d7bdaa9a8aad789af5641325ba31d43f3ba84b6b
[root@ip-172-31-89-80 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
a164990fb54f   sonarqube "/opt/sonarqube/dock..." 27 seconds ago Up 25 seconds 0.0.0.0:9000->9000/tcp, :::9000->9000/tcp flamboyant_fermat
[root@ip-172-31-89-80 ~]#
```

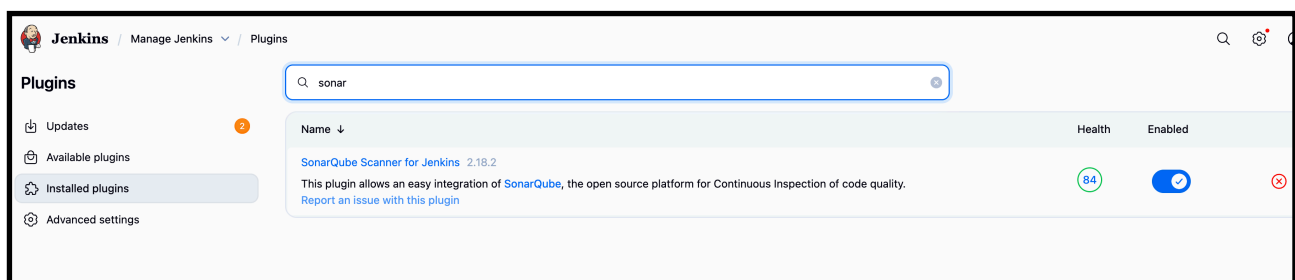
Step 2

Install sonrQube Scanner Plugin in Jenkins

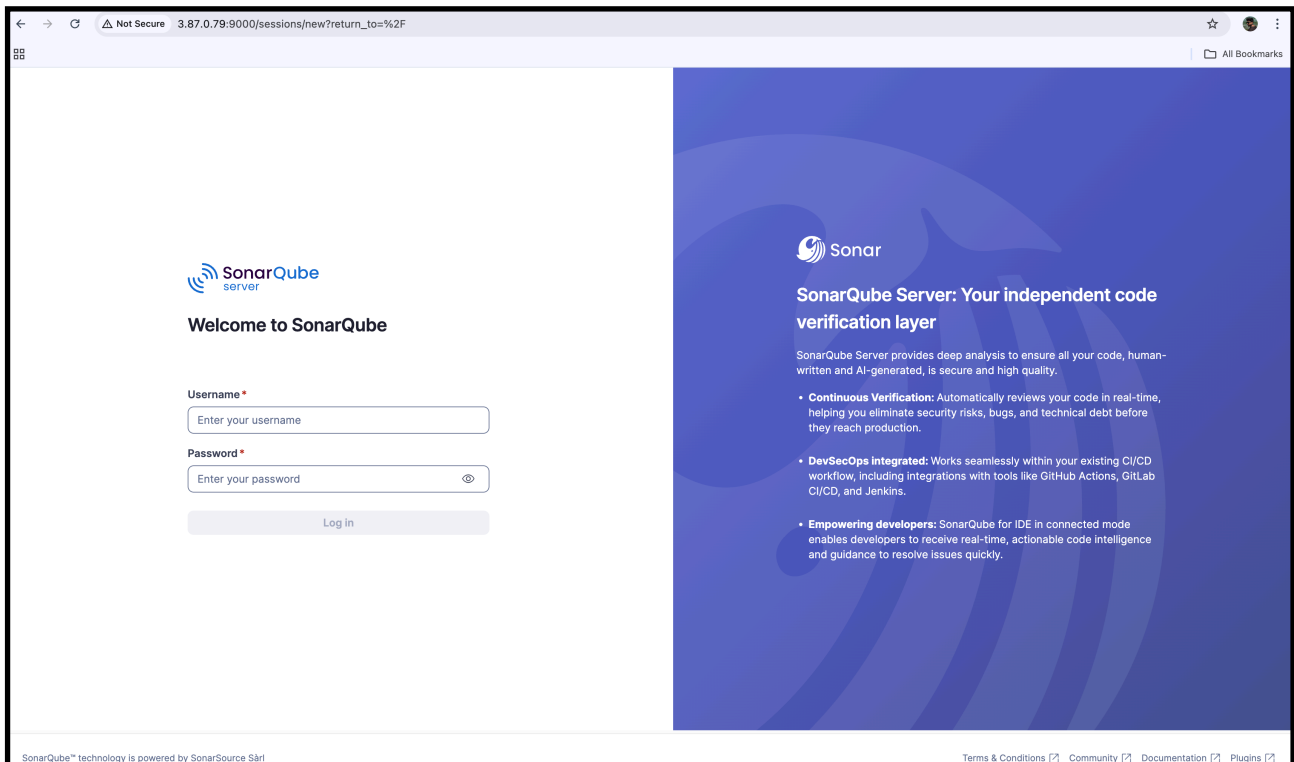
In Jenkins Dashboard:

- Go to **Manage Jenkins → Plugins → Available Plugins**.
- Search & install:
- **SonarQube Scanner for Jenkins**

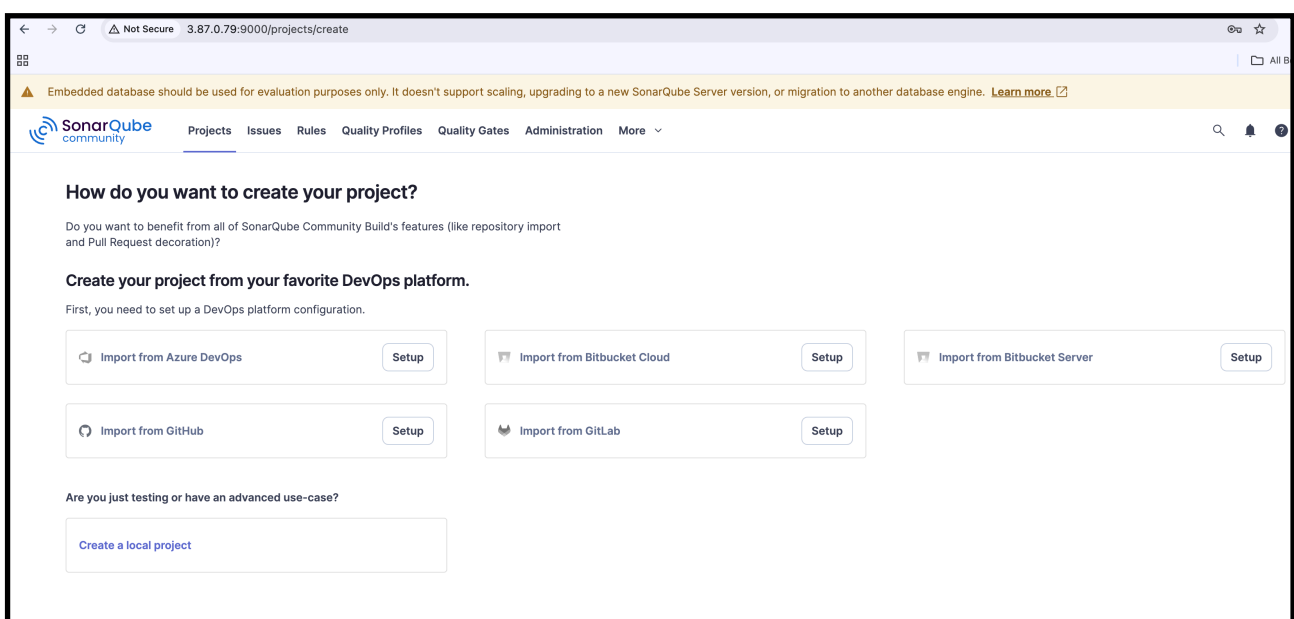
→ I already downloaded it



- > now login to your sonarqube
- > add the id password
- > and later give your own password and save



- > after login go to the security



—-> and generate a token

The screenshot shows the SonarQube Administrator interface. The top navigation bar includes 'Projects', 'Issues', 'Rules', 'Quality Profiles', 'Quality Gates', 'Administration', and 'More'. The user is logged in as 'Administrator'. The 'Security' tab is selected in the sub-navigation. The 'Security' section contains a paragraph explaining the purpose of User Tokens. Below this is the 'Generate Tokens' section with a form containing three input fields: 'Name' (with placeholder 'Enter Token Name'), 'Type' (with a dropdown menu 'Select Token Type'), and 'Expires in' (with a dropdown menu '30 days'). A 'Generate' button is to the right of these fields. Below the form is a table with columns: 'Name', 'Type', 'Project', 'Last use', 'Created', and 'Expiration'. The table is currently empty, showing 'No tokens'. At the bottom of the page, there is a section 'Enter a new password' with an 'Old Password' field.

—-> copy the token

Step 3:

- > now go to your Jenkins master
- > create new credentials
- > select the secret text
 - > in side secret paste the token from sonarqube
- > save

The screenshot shows the Jenkins 'New credentials' form. The breadcrumb navigation at the top is 'Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestrict...)'. The form has the following fields: 'Kind' (a dropdown menu set to 'Secret text'), 'Scope' (a dropdown menu with a help icon, set to 'Global (Jenkins, nodes, items, all child items, etc)'), 'Secret' (a text input field with masked characters), 'ID' (a text input field with a help icon, containing 'sonar-token'), and 'Description' (a text input field with a help icon, containing 'sonar-token'). A blue 'Create' button is at the bottom left of the form.

Configure SonarQube Server in Jenkins

Manage Jenkins

➔ **System**

➔ Scroll to **SonarQube servers**

➔ **Add SonarQube**

Fill:



The screenshot shows the 'Add SonarQube' configuration form in Jenkins. The form is titled 'Name' and contains the following fields:

- Name:** sonarqube-server
- Server URL:** http://98.93.150.46:9000 (Default is http://localhost:9000)
- Server authentication token:** slack2 (Mandatory when anonymous access is disabled)
- Advanced:** A dropdown menu with a downward arrow.

The form is enclosed in a dashed border with a red 'X' icon in the top right corner.

—-> after saving

—-> go to the job hiring-app

Configure Project for Sonar Analysis

Edit your Jenkins job where hiring-app Git repo is configured

➔ **Configure Job**

➔ Add a new build step: —>

sonar.projectKey=Sabear

sonar.projectName=Sabear

sonar.projectVersion=1.0

sonar.sources=/var/lib/jenkins/workspace/\$JOB_NAME/src/

sonar.binaries=target/classes/com/visualpathit/account/controller/

sonar.junit.reportsPath=target/surefire-reports

sonar.jacoco.reportPath=target/jacoco.exec

sonar.java.binaries=src/main/java/com/visualpathit/account/

JDK ?
JDK to be used for this SonarQube analysis
(Inherit From Job)

Path to project properties ?

Analysis properties ?
sonar.projectKey=Sabear
sonar.projectName=Sabear
sonar.projectVersion=1.0
sonar.sources=/var/lib/jenkins/workspace/\$JOB_NAME/src/
sonar.binaries=target/classes/com/visualpathit/account/controller/
sonar.junit.reportsPath=target/surefire-reports
sonar.jacoco.reportPath=target/jacoco.exec
sonar.java.binaries=src/main/java/com/visualpathit/account/

Additional arguments ?

JVM Options ?

Jenkins / hiring-app

Status

hiring-app

SonarQube

Permalinks

- Last build (#3), 2 hr 40 min ago
- Last stable build (#3), 2 hr 40 min ago
- Last successful build (#3), 2 hr 40 min ago
- Last failed build (#2), 2 hr 44 min ago
- Last unsuccessful build (#2), 2 hr 44 min ago
- Last completed build (#3), 2 hr 40 min ago

Builds

Filter

Today

- ✓ #3 11:59 AM
- ✗ #2 11:55 AM
- ✗ #1 11:54 AM

Build hiring-app Job

Click **Build Now**

Job should push code to SonarQube → You will see your project in Sonar UI

Delete Project

SonarQube

Rename

Builds

Filter

Today

#4 2:41 PM

- [Last stable build \(#3\), 2 hr 41 min ago](#)
- [Last successful build \(#3\), 2 hr 41 min ago](#)
- [Last failed build \(#2\), 2 hr 46 min ago](#)
- [Last unsuccessful build \(#2\), 2 hr 46 min ago](#)
- [Last completed build \(#3\), 2 hr 41 min ago](#)

—-> When done → Go to SonarQube dashboard → Check project results

1 project(s)

Perspective: Overall Status

Sort by: Name

Sabear

Passed

Last analysis: 42 seconds ago

Bugs

3

Vulnerabilities

0

Hotspots Reviewed

-

Code Smells

0

Coverage

-

Duplications

0.0%

Lines

11 XS XML, JSP

1 of 1 shown

C) --> Maven Compilation

Install & Configure Maven in Jenkins

download apache maven in jenkins master

```
cd /opt
```

```
wget https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
```

--> Ls check the tar file package of maven available

```
[root@ip-172-31-65-24 opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
--2025-12-31 08:45:01-- https://dlcdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9233336 (8.8M) [application/x-gzip]
Saving to: 'apache-maven-3.9.12-bin.tar.gz'

apache-maven-3.9.12-bin.tar.gz          100%[=====>] 8.80M  --.-KB/s  in 0.02s

2025-12-31 08:45:01 (384 MB/s) - 'apache-maven-3.9.12-bin.tar.gz' saved [9233336/9233336]

[root@ip-172-31-65-24 opt]#
```

--> tar -xvf apache-maven-3.9.12-bin.tar.gz

```
[root@ip-172-31-65-24 opt]# tar -xvf apache-maven-3.9.12-bin.tar.gz
apache-maven-3.9.12/README.txt
apache-maven-3.9.12/LICENSE
apache-maven-3.9.12/NOTICE
apache-maven-3.9.12/lib/
apache-maven-3.9.12/lib/aopalliance.license
apache-maven-3.9.12/lib/asm.license
apache-maven-3.9.12/lib/commons-cli.license
apache-maven-3.9.12/lib/commons-codec.license
apache-maven-3.9.12/lib/error_prone_annotations.license
apache-maven-3.9.12/lib/failureaccess.license
apache-maven-3.9.12/lib/gson.license
apache-maven-3.9.12/lib/guava.license
apache-maven-3.9.12/lib/guice.license
apache-maven-3.9.12/lib/httpclient.license
```

To extract the maven packages

```
[root@ip-172-31-65-24 opt]# ls
apache-maven-3.9.12  apache-maven-3.9.12-bin.tar.gz  aws
[root@ip-172-31-65-24 opt]#
```

```
cd apache-maven-3.9.12/bin
```

```
apache-maven-3.9.12  apache-maven-3.9.12-bin.tar.gz  aws
[root@ip-172-31-65-24 opt]# cd apache-maven-3.9.12/bin
[root@ip-172-31-65-24 bin]# ls
m2.conf  mvn  mvn.cmd  mvnDebug  mvnDebug.cmd  mvnyjp
[root@ip-172-31-65-24 bin]#
```

--> copy the path pwd

```
[root@ip-172-31-95-0 bin]# pwd
/opt/apache-maven-3.9.12/bin
[root@ip-172-31-95-0 bin]# cd
[root@ip-172-31-95-0 ~]# ls
[root@ip-172-31-95-0 ~]# vi .basah_profile
[root@ip-172-31-95-0 ~]# vi .bash_profile
[root@ip-172-31-95-0 ~]# mvn --version
-bash: mvn: command not found
[root@ip-172-31-95-0 ~]# vi .bash_profile
[root@ip-172-31-95-0 ~]#
```

--> edit the .bash_profile

--> vi .bash_profile

Below the # user specific environment

```
MVN_HOME=/opt/apache-maven-3.9.12/bin
```

```
PATH=$PATH:$HOME/bin:$MVN_HOME
```



```
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs
MVN_HOME=/opt/apache-maven-3.9.12/bin
PATH=$PATH:$HOME/bin:$MVN_HOME

export PATH
```

export PATH

~

—> save and exit

Next

—> source ~/.bash_profile

—> echo \$MVN_HOME

—> mvn --version

```
[root@ip-172-31-65-24 ~]# mvn --version
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9a7a1)
Maven home: /opt/apache-maven-3.9.12
Java version: 17.0.17, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.158-180.294.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-65-24 ~]#
```

```
[root@ip-172-31-65-24 bin]# cd
[root@ip-172-31-65-24 ~]# vi .bash_profile
[root@ip-172-31-65-24 ~]# source ~/.bash_profile
[root@ip-172-31-65-24 ~]# echo $MVN_HOME
/opt/apache-maven-3.9.12/bin
```

Now go to manage jenkins --> tools --> maven --> /opt/apache-maven-3.9.12

 **Jenkins** / Manage Jenkins ▾ / Tools

Gradle installations

+ Add Gradle

SonarScanner for MSBuild installations

+ Add SonarScanner for MSBuild

SonarQube Scanner installations

SonarQube Scanner installations ▾ [Edited](#)

Ant installations

+ Add Ant

Maven installations

+ Add Maven

Save

Apply

 **Maven**

Name

MVN_HOME

MAVEN_HOME

/opt/apache-maven-3.9.12

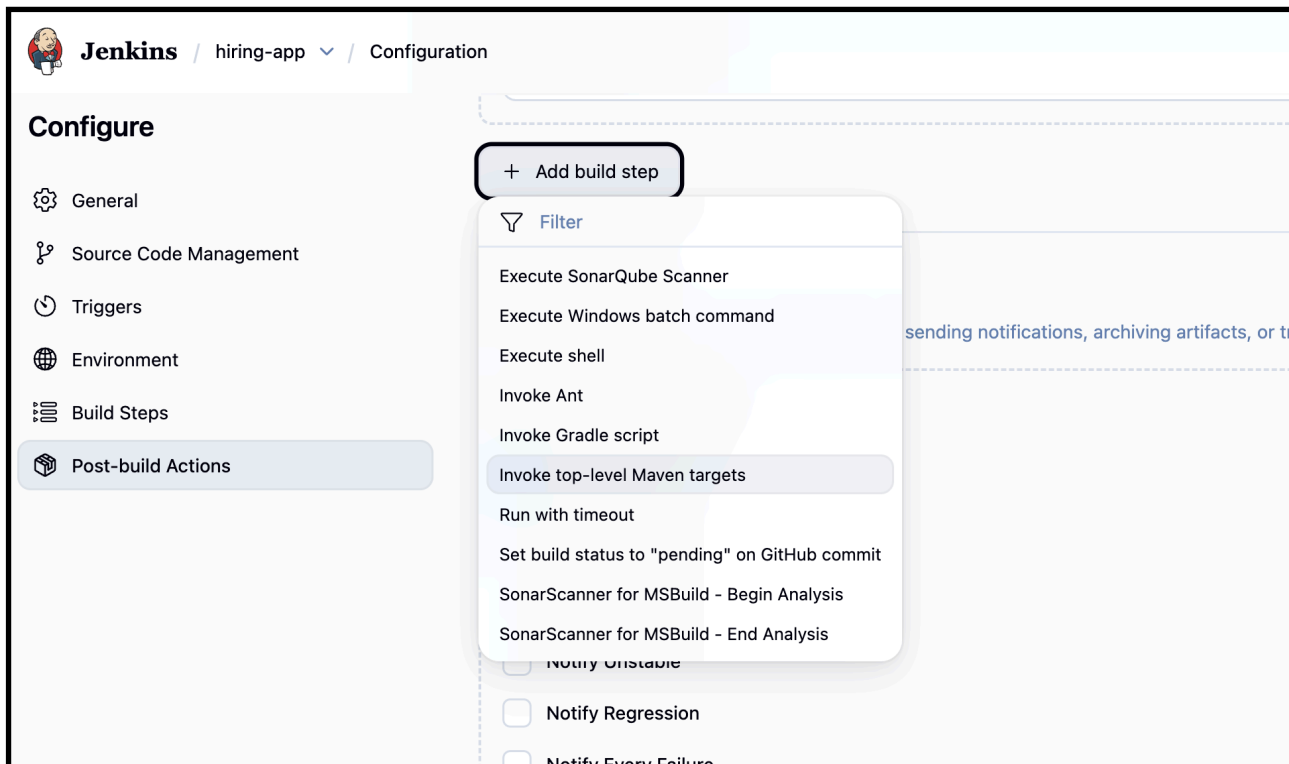
☐ Install automatically

?

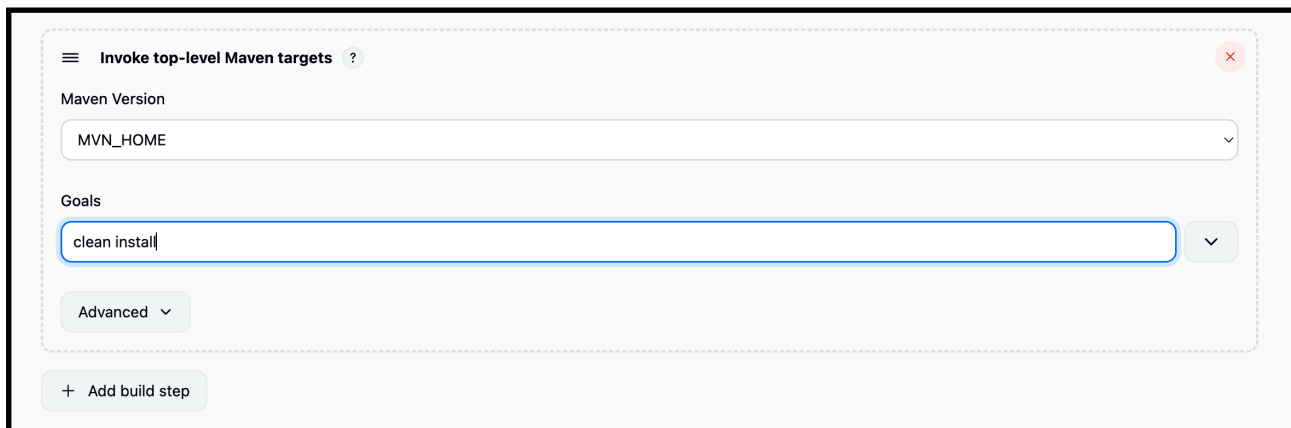
+ Add Maven

Add build step → **Invoke top-level Maven targets.**

Goals: clean



The screenshot shows the Jenkins Configuration page for a job named 'hiring-app'. The left sidebar contains a 'Configure' section with various tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'Post-build Actions' tab is selected. A dropdown menu is open, showing a list of build steps. The step 'Invoke top-level Maven targets' is highlighted. Other visible steps include 'Execute SonarQube Scanner', 'Execute Windows batch command', 'Execute shell', 'Invoke Ant', 'Invoke Gradle script', 'Run with timeout', 'Set build status to "pending" on GitHub commit', 'SonarScanner for MSBuild - Begin Analysis', and 'SonarScanner for MSBuild - End Analysis'.



The screenshot shows the configuration for the 'Invoke top-level Maven targets' build step. The 'Maven Version' dropdown is set to 'MVN_HOME'. The 'Goals' text input field contains 'clean install'. Below the input fields is an 'Advanced' dropdown menu. At the bottom of the configuration area is a '+ Add build step' button.

—> save

Now run the build then git clone done, sonarqube integration done, then maven created a war file.

The screenshot shows the Jenkins web interface for a build named 'hiring-app'. The 'Console Output' tab is selected, displaying a log of the build process. The log starts with 'Started by user waseem akram' and 'Running as SYSTEM'. It shows the build directory as '/var/lib/jenkins/workspace/hiring-app'. The log includes git commands for cloning the repository, fetching changes, and checking out a specific revision. It also shows the SonarScanner configuration and the execution of the scanner. The log ends with the user cache path: '/var/lib/jenkins/.sonar/cache'.

```
Started by user waseem akram
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/hiring-app
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/hiring-app/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953ealf9669784f9d6a (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9405b107fdffdf68e69fe953ealf9669784f9d6a # timeout=10
Commit message: "Update README.md"
> git rev-list --no-walk 9405b107fdffdf68e69fe953ealf9669784f9d6a # timeout=10
[hiring-app] $ /var/lib/jenkins/tools/hudson.plugins.sonar.SonarRunnerInstallation/SonarScanner/bin/sonar-scanner -
Dsonar.host.url=http://13.222.252.108:9000/ ***** -Dsonar.projectKey=Sabear -Dsonar.projectName=Sabear -Dsonar.jacoco.report
Dsonar.projectVersion=1.0 -Dsonar.sources=/var/lib/jenkins/workspace/hiring-app/src/ -Dsonar.java.binaries=src/main/java/com/vi
Dsonar.binaries=target/classes/com/visualpathit/account/controller/ -Dsonar.junit.reportsPath=target/surefire-reports -
Dsonar.projectBaseDir=/var/lib/jenkins/workspace/hiring-app
08:54:17.140 INFO Scanner configuration file: /var/lib/jenkins/tools/hudson.plugins.sonar.SonarRunnerInstallation/SonarScanner
scanner.properties
08:54:17.146 INFO Project root configuration file: NONE
08:54:17.160 INFO SonarScanner CLI 8.0.1.6346
08:54:17.162 INFO Linux 6.1.158-180.294.amzn2023.x86_64 amd64
08:54:18.434 INFO Communicating with SonarQube Server 9.9.1.69595
08:54:18.956 INFO Load global settings
08:54:19.016 INFO Load global settings (done) | time=62ms
08:54:19.017 INFO Server id: 92D88F0A-AZtqf_Xvo47LHJ5dS0Fg
08:54:19.021 INFO User cache: /var/lib/jenkins/.sonar/cache
```

In jenkins server → `cd /var/lib/jenkins/workspace/hiring-app/target`

```
OS name: Linux, version: 6.1.158-180.294.amzn2023.x86_64, arch: amd64, family:
[root@ip-172-31-65-24 ~]# cd /var/lib/jenkins/workspace/hiring-app/target
[root@ip-172-31-65-24 target]# ls
hiring hiring.war maven-archiver
[root@ip-172-31-65-24 target]#
```

`cd /var/lib/jenkins/workspace/hiring-app/target`

—-> now check the `cd /var/lib/jenkins/.m2/repository`

```
cd /var/lib/jenkins/.m2/repository
```

```
[root@ip-172-31-65-24 jenkins]# cd .m2
[root@ip-172-31-65-24 .m2]# ls
repository
[root@ip-172-31-65-24 .m2]# cd repository/
[root@ip-172-31-65-24 repository]# ls
aopalliance  com  commons-io  in  javax  junit  net  org
[root@ip-172-31-65-24 repository]# pwd
/var/lib/jenkins/.m2/repository
[root@ip-172-31-65-24 repository]#
```

d). Nexus Artifactory

1. Now create nexus-ec2 with t2.large
2. Install java → **yum -y install java-1.8***

```
#  
- \ ##\ Amazon Linux 2023  
-- \ #####\  
   \|#####|  
    \###|  
     \#/ https://aws.amazon.com/linux/amazon-linux-2023  
      v~' '->  
  
-----  
---.  
_/_/\_-/  
-/m/' -/
```

```
[ec2-user@ip-172-31-87-107 ~]$ sudo -i  
[root@ip-172-31-87-107 ~]# sudo yum update -y  
Waiting for process with pid 2406 to finish.  
Amazon Linux 2023 Kernel Livepatch repository  
Last metadata expiration check: 0:00:01 ago on Wed Dec 31 09:04:09 2025.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@ip-172-31-87-107 ~]# yum -y install java-1.8.*  
Last metadata expiration check: 0:00:20 ago on Wed Dec 31 09:04:09 2025.  
Dependencies resolved.
```

Package	Architecture	Version
Installing:		
java-1.8.0-amazon-corretto	x86_64	1:1.8.0_472
java-1.8.0-amazon-corretto-devel	x86_64	1:1.8.0_472
Installing dependencies:		
adwaita-cursor-theme	noarch	47.0-1.amzn
adwaita-icon-theme	noarch	47.0-1.amzn
adwaita-icon-theme-legacy	noarch	46.2-2.amzn
alsa-lib	x86_64	1.2.7.2-1.ar
at-spi2-atk	x86_64	2.54.0-1.am
at-spi2-core	x86_64	2.54.0-1.am
atk	x86_64	2.54.0-1.am
avahi-glib	x86_64	0.8-14.amzn

1. And now download nexus

```
cd /opt
```

```
sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz
```

```
[root@ip-172-31-94-110 opt]# sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz
--2026-01-02 07:52:20-- https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz
Resolving download.sonatype.com (download.sonatype.com)... 54.215.226.116, 54.241.7.25
Connecting to download.sonatype.com (download.sonatype.com)|54.215.226.116|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.68.1-02-unix.tar.gz [following]
--2026-01-02 07:52:20-- https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.68.1-02-unix.tar.gz
Resolving cdn.download.sonatype.com (cdn.download.sonatype.com)... 104.18.0.134, 104.18.1.134, 2606:4700::6812:86, ...
Connecting to cdn.download.sonatype.com (cdn.download.sonatype.com)|104.18.0.134|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 228724143 (218M) [application/x-gzip]
Saving to: 'nexus-3.68.1-02-unix.tar.gz'

nexus-3.68.1-02-unix.tar.gz      100%[=====>] 218.13M  342
2026-01-02 07:52:21 (342 MB/s) - 'nexus-3.68.1-02-unix.tar.gz' saved [228724143/228724143]

[root@ip-172-31-94-110 opt]#
```

```
sudo tar -xvf nexus-3.68.1-02-unix.tar.gz
```

```
2026-01-02 07:52:21 (342 MB/s) - 'nexus-3.68.1-02-unix.tar.gz' saved [228724143/228724143]

[root@ip-172-31-94-110 opt]# ls
aws  nexus-3.68.1-02-unix.tar.gz
[root@ip-172-31-94-110 opt]# sudo tar -xvf nexus-3.68.1-02-unix.tar.gz
nexus-3.68.1-02/.install4j/9d17dc87.lprop
nexus-3.68.1-02/.install4j/MessagesDefault
nexus-3.68.1-02/.install4j/build.uuid
nexus-3.68.1-02/.install4j/e4ada6b7.lprop
nexus-3.68.1-02/.install4j/i4j_extf_0_17islik.utf8
nexus-3.68.1-02/.install4j/i4j_extf_10_17islik_10358jn.png
nexus-3.68.1-02/.install4j/i4j_extf_11_17islik_lgne9sv.png
nexus-3.68.1-02/.install4j/i4j_extf_12_17islik_sc8j43.png
nexus-3.68.1-02/.install4j/i4j_extf_13_17islik_10nxrsm.png
nexus-3.68.1-02/.install4j/i4j_extf_14_17islik_yd7am4.png
nexus-3.68.1-02/.install4j/i4j_extf_15_17islik_10nxrsm.png
```

```
sudo mv nexus-3.68.1-02 nexus
```

```
sonatype-work/nexus3/tmp/.placeholder
[root@ip-172-31-94-110 opt]# ll
total 223364
drwxr-xr-x.  4 root root      33 Dec  3 16:42 aws
drwxr-xr-x. 10 root root     181 Jan  2 07:53 nexus-3.68.1-02
-rw-r--r--.  1 root root 228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz
drwxr-xr-x.  3 root root      20 Jan  2 07:53 sonatype-work
[root@ip-172-31-94-110 opt]#
```

```
total 223364
drwxr-xr-x.  4 root root      33 Dec  3 16:42 aws
drwxr-xr-x. 10 root root     181 Jan  2 07:53 nexus-3.68.1-02
-rw-r--r--.  1 root root 228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz
drwxr-xr-x.  3 root root      20 Jan  2 07:53 sonatype-work
[root@ip-172-31-94-110 opt]# sudo mv nexus-3.68.1-02 nexus
[root@ip-172-31-94-110 opt]# ll
total 223364
drwxr-xr-x.  4 root root      33 Dec  3 16:42 aws
drwxr-xr-x. 10 root root     181 Jan  2 07:53 nexus
-rw-r--r--.  1 root root 228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz
drwxr-xr-x.  3 root root      20 Jan  2 07:53 sonatype-work
[root@ip-172-31-94-110 opt]#
```

Open /opt/nexus/bin/nexus.rc file,
uncomment run_as_user parameter and set it as following.
vi /opt/nexus/bin/nexus.rc
Remove # and Add → run_as_user="nexus"

```
run_as_user="nexus"
```

```
[root@ip-172-31-87-107 opt]# cd  
[root@ip-172-31-87-107 ~]# vi /opt/nexus/bin/nexus.rc  
[root@ip-172-31-87-107 ~]#
```

sudo useradd nexus
sudo chown -R nexus:nexus /opt/nexus /opt/sonatype-work (to change owner ship to user nexus)

```
drwxr-xr-x. 4 root root      33 Dec  3 16:42 aws  
drwxr-xr-x. 10 root root    181 Jan  2 07:53 nexus  
-rw-r--r--. 1 root root 228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz  
drwxr-xr-x. 3 root root      20 Jan  2 07:53 sonatype-work  
[root@ip-172-31-94-110 opt]# sudo useradd nexus  
useradd: user 'nexus' already exists  
[root@ip-172-31-94-110 opt]# sudo chown -R nexus:nexus /opt/nexus /opt/sonatype-work  
[root@ip-172-31-94-110 opt]# ll  
total 223364  
drwxr-xr-x. 4 root root      33 Dec  3 16:42 aws  
drwxr-xr-x. 10 nexus nexus    181 Jan  2 07:53 nexus  
-rw-r--r--. 1 root root 228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz  
drwxr-xr-x. 3 nexus nexus      20 Jan  2 07:53 sonatype-work  
[root@ip-172-31-94-110 opt]#
```

- Cd /opt/nexus/bin
- ./nexus start
- ./nexus status

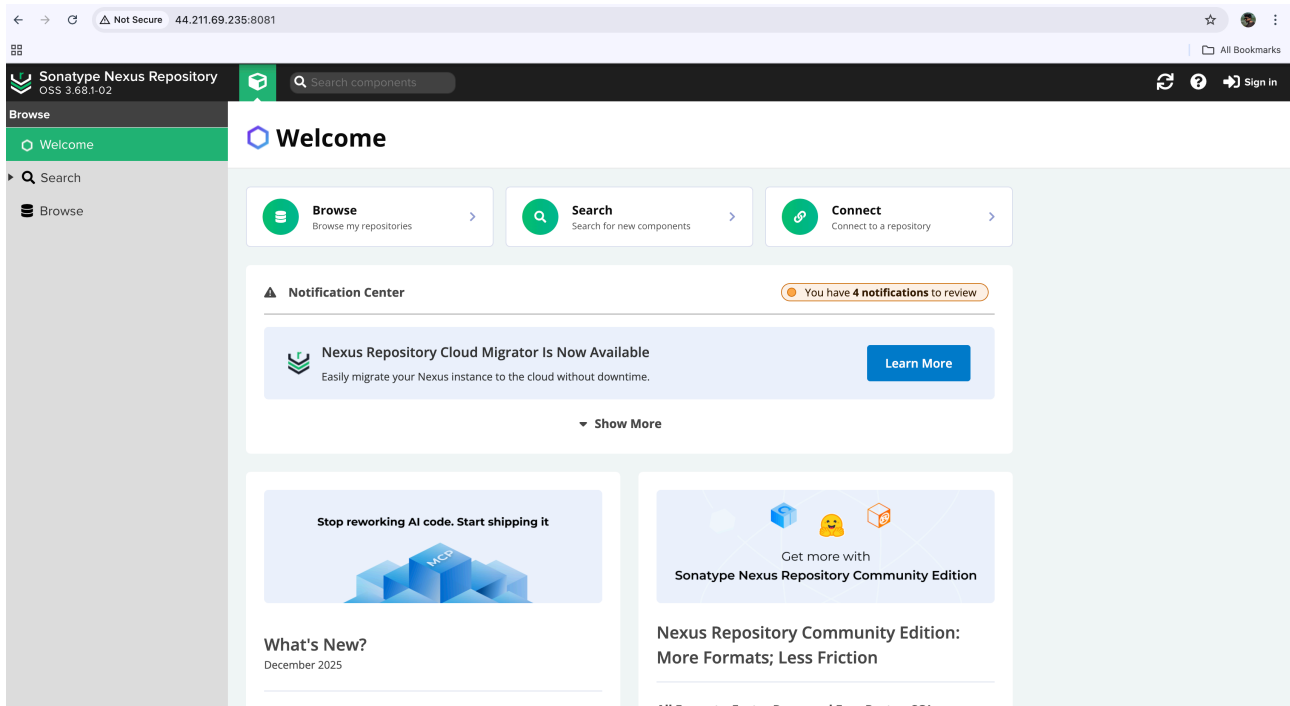
```
[sudo] password for nexus:
sudo: a password is required
[nexus@ip-172-31-94-110 ~]$ cd /opt/nexus/bin
[nexus@ip-172-31-94-110 bin]$ ./nexus start
Starting nexus
[nexus@ip-172-31-94-110 bin]$ ./nexus status
nexus is running.
[nexus@ip-172-31-94-110 bin]$
```

i-0dc2978b8f9efc9f4 (nexus)

PublicIPs: 44.211.69.235 PrivateIPs: 172.31.94.110

Let's install Nexus Repository OSS 3.68.1-02 for Linux	
•	<code>cd /opt</code>
•	<code>sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz</code>
•	<code>sudo tar -xvf nexus-3.68.1-02-unix.tar.gz</code>
•	<code>sudo mv nexus-3.68.1-02 nexus</code>
•	<code>sudo useradd nexus</code>
•	<code>sudo chown -R nexus:nexus /opt/nexus /opt/sonatype-work</code>

—> now go and check the ip port <public ip> 8081
Nexus default port is 8081



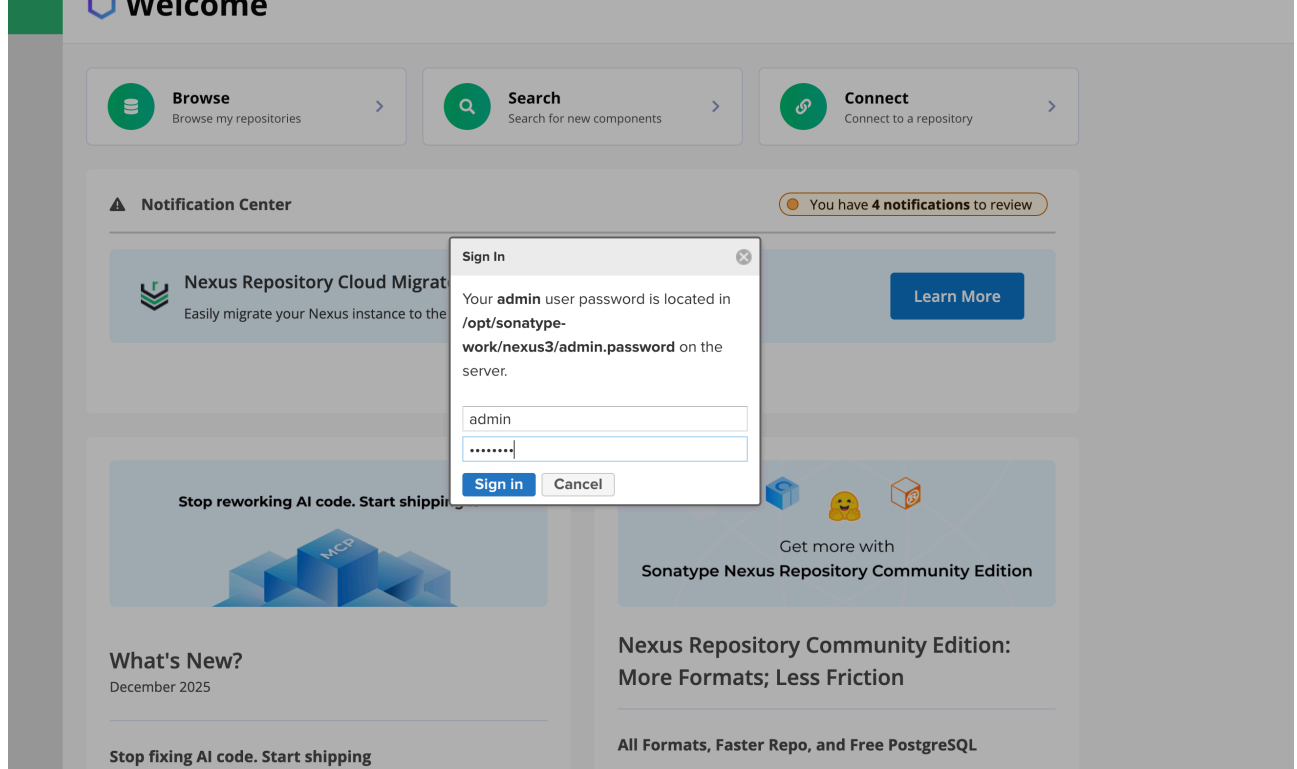
Sign In : admin

Password : 329be116-db9a-4fac-bd01-553cf0db2c8e

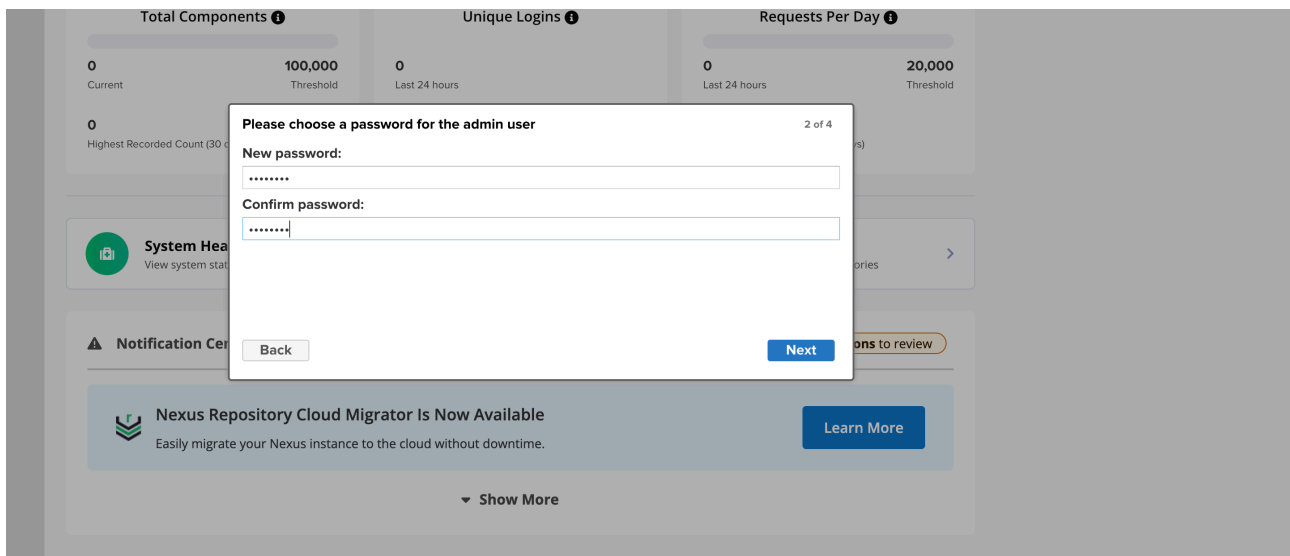
—> cat /opt/sonatype-work/nexus3/admin.password

Copy the password and paste in nexus login password

```
-rw-r--r--. 1 nexus nexus 1635 May 16 2024 nexus.vmoptions
[nexus@ip-172-31-94-110 bin]$ cat /opt/sonatype-work/nexus3/admin.password
329be116-db9a-4fac-bd01-553cf0db2c8e[nexus@ip-172-31-94-110 bin]$
```

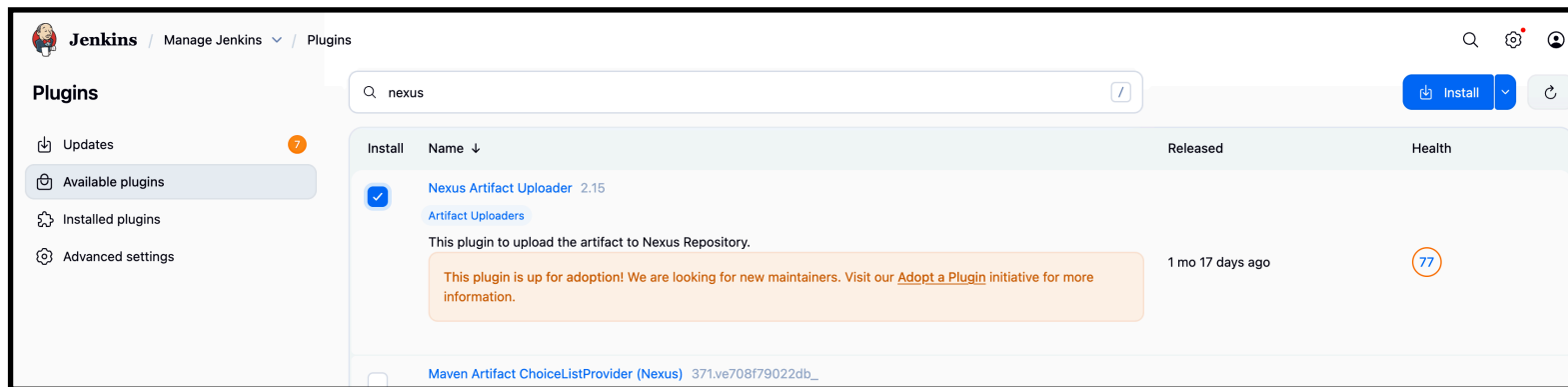


—> after login change the password as you require



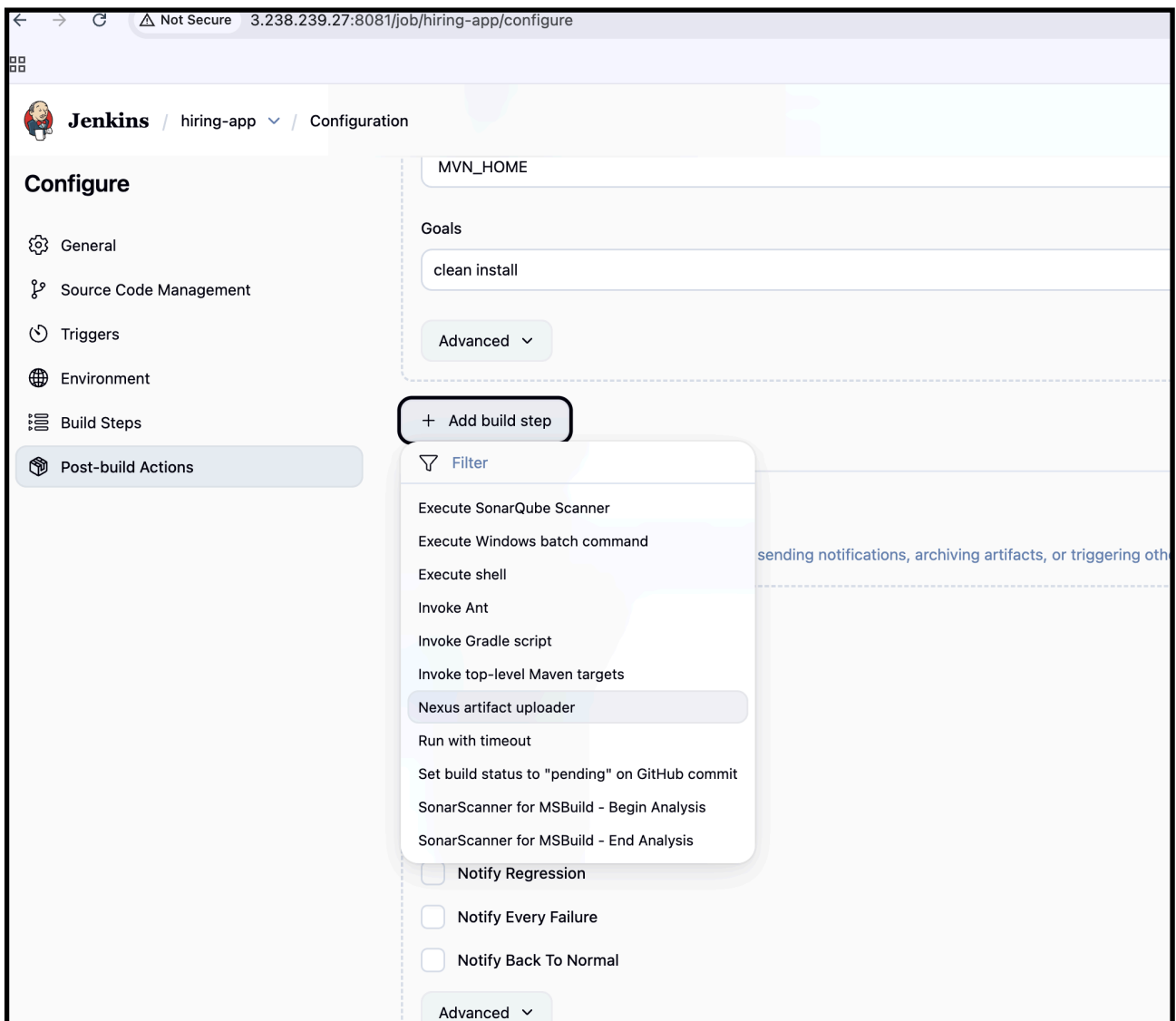
Integrate jenkins with nexus

Install Nexus plugin



Now go to your job → configure → build step → nexus

In git hub repo in pom.xml file we have some info group id and version, bring and enter here and goto nexus create repository → maven2(host) and take release.



hiring-app / Configuration

management

ns

Nexus artifact uploader

Nexus Details

Nexus Version
NEXUS2

Protocol
HTTP

Nexus URL ?
44.211.69.235:8081

Credentials
- none - + Add

Groupid

❗ Groupid must not be empty

Version

❗ Version must not be empty

—> now create a credential

The image shows a screenshot of the Jenkins web interface, specifically the 'Jenkins Credentials Provider: Jenkins' configuration dialog. The dialog is titled 'Jenkins Credentials Provider: Jenkins' and contains several fields for configuring a credential. The 'Kind' dropdown is set to 'Username with password'. The 'Scope' dropdown is set to 'Global (Jenkins, nodes, items, all child items, etc)'. The 'Username' field contains the text 'admin'. There is an unchecked checkbox labeled 'Treat username as secret'. The 'Password' field is masked with dots. The 'ID' field contains the text 'nexus-user'. The 'Description' field contains the text 'nexus-user'. At the bottom right of the dialog are 'Cancel' and 'Add' buttons. The background shows a blurred view of the Jenkins interface with a sidebar on the right.

Jenkins Credentials Provider: Jenkins

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

admin

☐ Treat username as secret ?

Password ?

.....

ID ?

nexus-user

Description ?

nexus-user

Cancel Add

--> attach that credential

--> now go to GitHub repo of hiring app and open pom.file
And add fill the details

betawins / hiring-app

<> Code Pull requests Actions Projects Security Insights

main hiring-app / pom.xml

bcreddydevops Update pom.xml

Code Blame 28 lines (28 loc) · 890 Bytes

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>in.javahome</groupId>
5   <artifactId>hiring</artifactId>
6   <packaging>war</packaging>
7   <version>0.1</version>
8   <name>hiring Maven Webapp</name>
9   <url>http://maven.apache.org</url>
10  <dependencies>
11    <dependency>
12      <groupId>junit</groupId>
13      <artifactId>junit</artifactId>
14      <version>3.8.1</version>
15      <scope>test</scope>
16    </dependency>
17  </dependencies>
18  <build>
19    <plugins>
20      <plugin>
21        <groupId>org.apache.maven.plugins</groupId>
22        <artifactId>maven-war-plugin</artifactId>
23        <version>3.3.1</version>
24      </plugin>
25    </plugins>
26    <finalName>hiring</finalName>
27  </build>
28 </project>
```

—-> now go to your Job configuration and add above details.
From repo

Nexus Details

Nexus Version

NEXUS3

Protocol

HTTP

Nexus URL ?

44.211.69.235:8081

Credentials

admin/***** (nexus-user)

+ Add

GroupId

in.javahome

Version

0.1

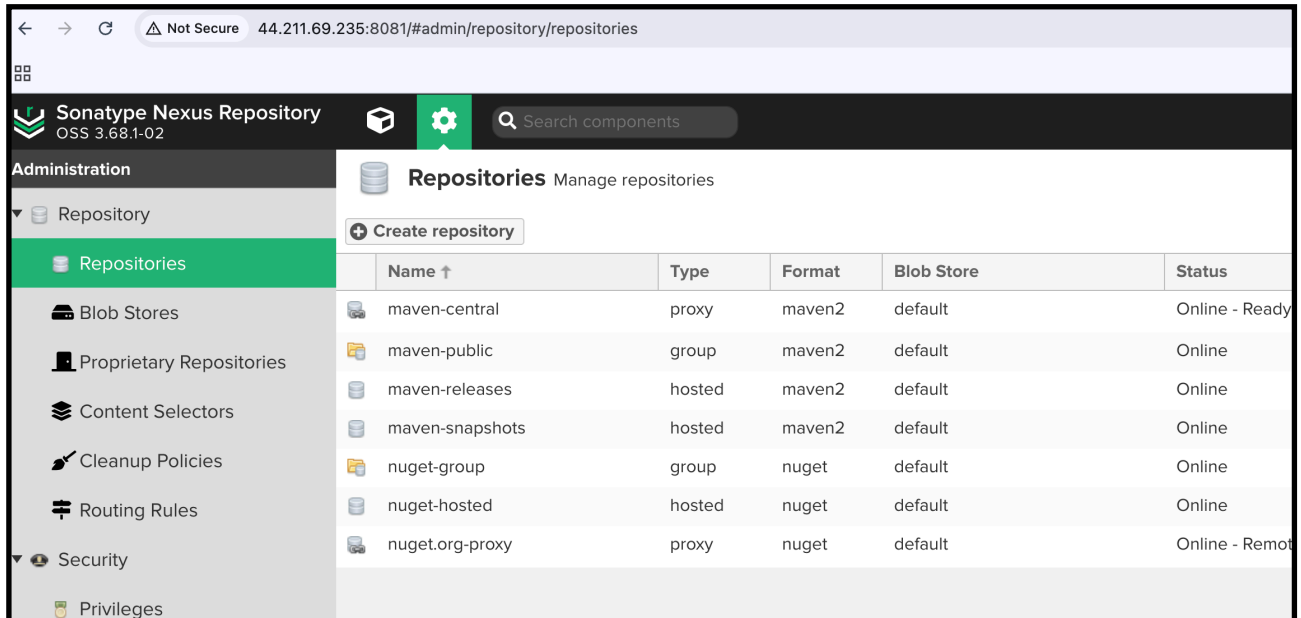
Repository ?

hiring-app

Artifacts

Artifact

—> now before adding repository name go to your nexus and create a repository



Sonatype Nexus Repository OSS 3.68.1-02

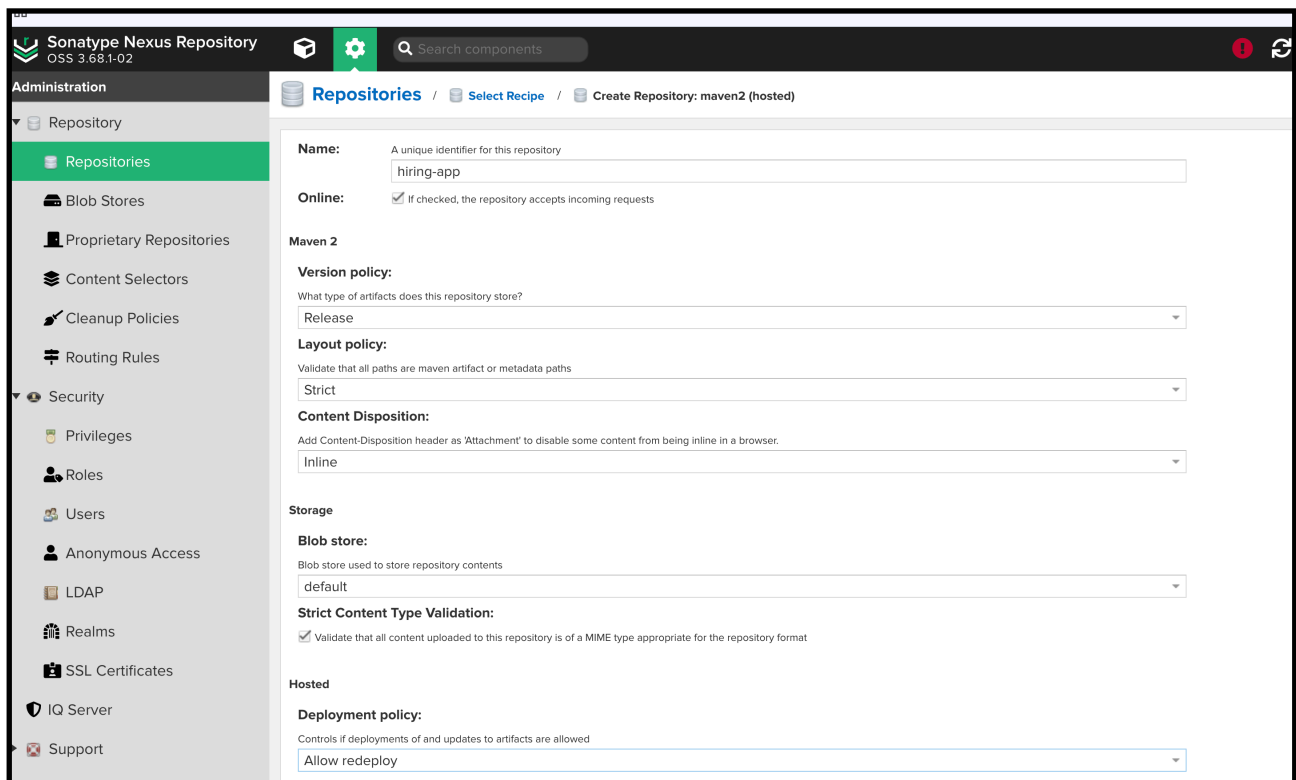
Administration

- Repository
- Repositories**
- Blob Stores
- Proprietary Repositories
- Content Selectors
- Cleanup Policies
- Routing Rules
- Security
- Privileges

Repositories Manage repositories

+ Create repository

Name ↑	Type	Format	Blob Store	Status
maven-central	proxy	maven2	default	Online - Ready
maven-public	group	maven2	default	Online
maven-releases	hosted	maven2	default	Online
maven-snapshots	hosted	maven2	default	Online
nuget-group	group	nuget	default	Online
nuget-hosted	hosted	nuget	default	Online
nuget.org-proxy	proxy	nuget	default	Online - Remote



Sonatype Nexus Repository OSS 3.68.1-02

Administration

- Repository
- Repositories**
- Blob Stores
- Proprietary Repositories
- Content Selectors
- Cleanup Policies
- Routing Rules
- Security
- Privileges
- Roles
- Users
- Anonymous Access
- LDAP
- Realms
- SSL Certificates
- IQ Server
- Support

Repositories / Select Recipe / Create Repository: maven2 (hosted)

Name: A unique identifier for this repository
hiring-app

Online: ☒ If checked, the repository accepts incoming requests

Maven 2

Version policy:
What type of artifacts does this repository store?
Release

Layout policy:
Validate that all paths are maven artifact or metadata paths
Strict

Content Disposition:
Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser.
Inline

Storage

Blob store:
Blob store used to store repository contents
default

Strict Content Type Validation:
☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy:
Controls if deployments of and updates to artifacts are allowed
Allow redeploy

—> select maven2 (hosted)
—> given name hiring-app

—> and create repository

Repository	Create repository							Filter
Repositories	Name ↑	Type	Format	Blob Store	Status	URL	Health check	
Blob Stores	hiring-app	hosted	maven2	default	Online	copy		

—-> now again go to same job configuration page and add the repository name

Nexus Details

Nexus Version

NEXUS3

Protocol

HTTP

Nexus URL ?

44.211.69.235:8081

Credentials

admin/***** (nexus-user)

+ Add

Groupid

in.javahome

Version

0.1

Repository ?

hiring-app

Artifacts

Artifact

Artifacts

Artifact

ArtifactId

hiring

Type ?

war

Classifier ?

File ?

target/hiring.war

+ Add

--> click save

Now run build

Jenkins hiring-app

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

SonarQube

Rename

BUILDS

Builds

Filter

Today

#15 9:56 AM

hiring-app

SonarQube

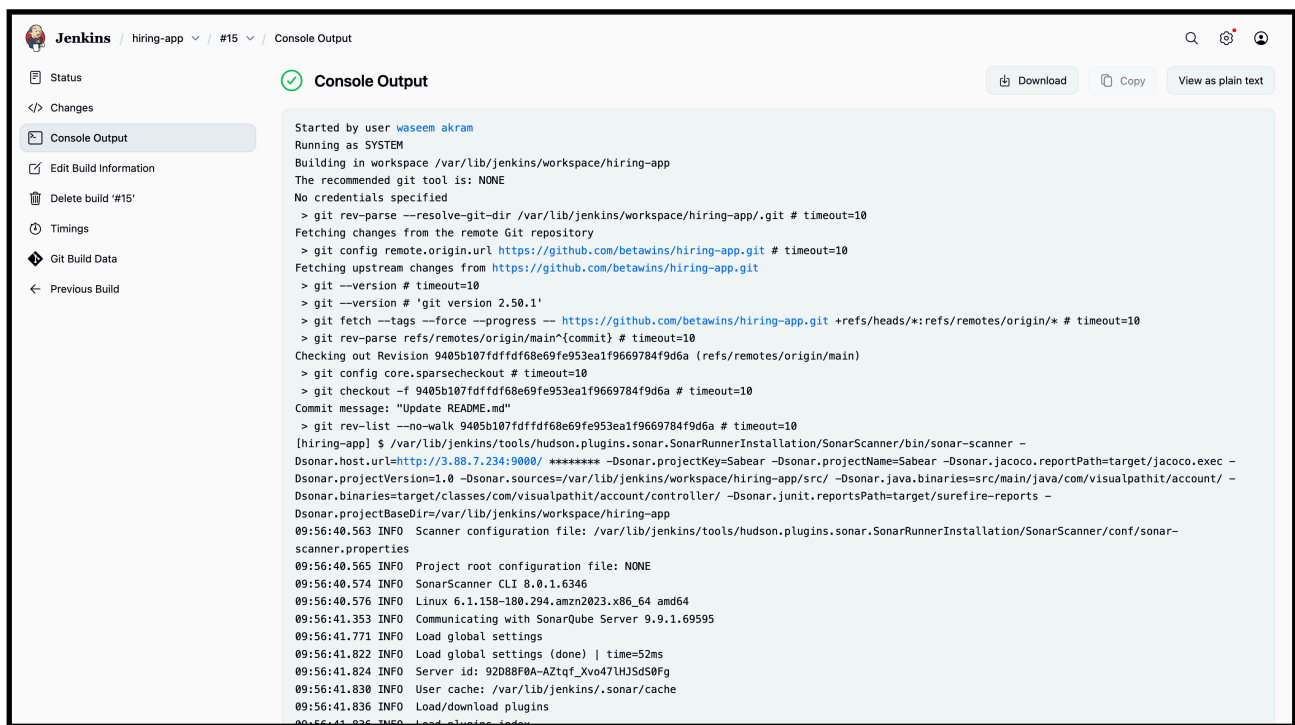
SonarQube Quality Gate

Sabear Passed

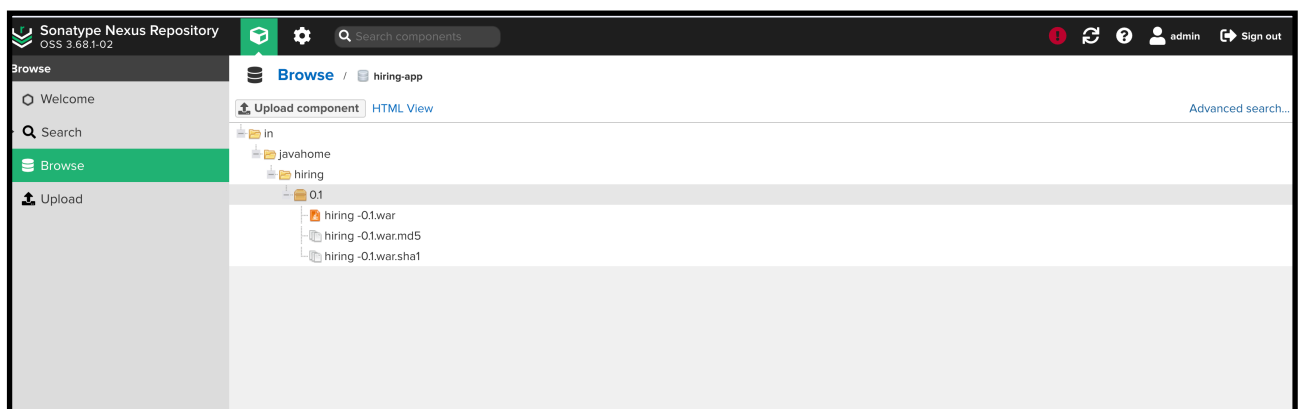
server-side processing: Success

Permalinks

- Last build (#15), 21 sec ago
- Last stable build (#15), 21 sec ago
- Last successful build (#15), 21 sec ago
- Last failed build (#14), 4 min 39 sec ago
- Last unsuccessful build (#14), 4 min 39 sec ago
- Last completed build (#15), 21 sec ago



Check Nexus for war file



E) slack notification

Slack

Workspace [?](#)

techiehorizon

Credential [?](#)

slack [▼](#) [+ Add](#)

Default channel / member id [?](#)

☐ Custom slack app bot user [?](#)

Advanced [▼](#)

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Slack Notifications

- ☒ Notify Build Start
- ☒ Notify Success
- ☐ Notify Aborted
- ☐ Notify Not Built
- ☐ Notify Unstable
- ☐ Notify Regression
- ☐ Notify Every Failure
- ☐ Notify Back To Normal

Advanced [▼](#)

jenkins

APP 3:22 PM

hiring-app - #14 Started by user waseem akram (Open)

hiring-app - #15 Started by user waseem akram (Open)

hiring-app - #15 Success after 7.7 sec (Open)

F) Deploy On tomcat

Go to jenkins page Install plugin → deploy to container

Jenkins

Manage Jenkins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Search

deploy

Install

7

Install	Name	Released	Health
☒	Deploy to container 1.17 Artifact Uploaders This plugin allows you to deploy a war to a container after a successful build. Glassfish 3.x remote deployment	7 mo 24 days ago	87
☐	Docker Pipeline 634.vedc7242b_sda_7 pipeline DevOps Deployment docker Build and use Docker containers from pipelines.		

—-> Now launch an ec2 instance
Install java and install tomcat in it

And start

```

complete:
[root@ip-172-31-10-35 ~]# sudo yum install -y java-17-amazon-corretto
Last metadata expiration check: 0:02:55 ago on Fri Jan  2 10:20:08 2026.
Dependencies resolved.
=====
Package                                Architecture          Version
=====
Installing:

```

```
[root@ip-172-31-10-35 ~]# cd /opt
[root@ip-172-31-10-35 opt]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz
--2026-01-02 10:25:05-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13049663 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.113.tar.gz'

apache-tomcat-9.0.113.tar.gz          100%[=====]
2026-01-02 10:25:05 (316 MB/s) - 'apache-tomcat-9.0.113.tar.gz' saved [13049663/13049663]
```

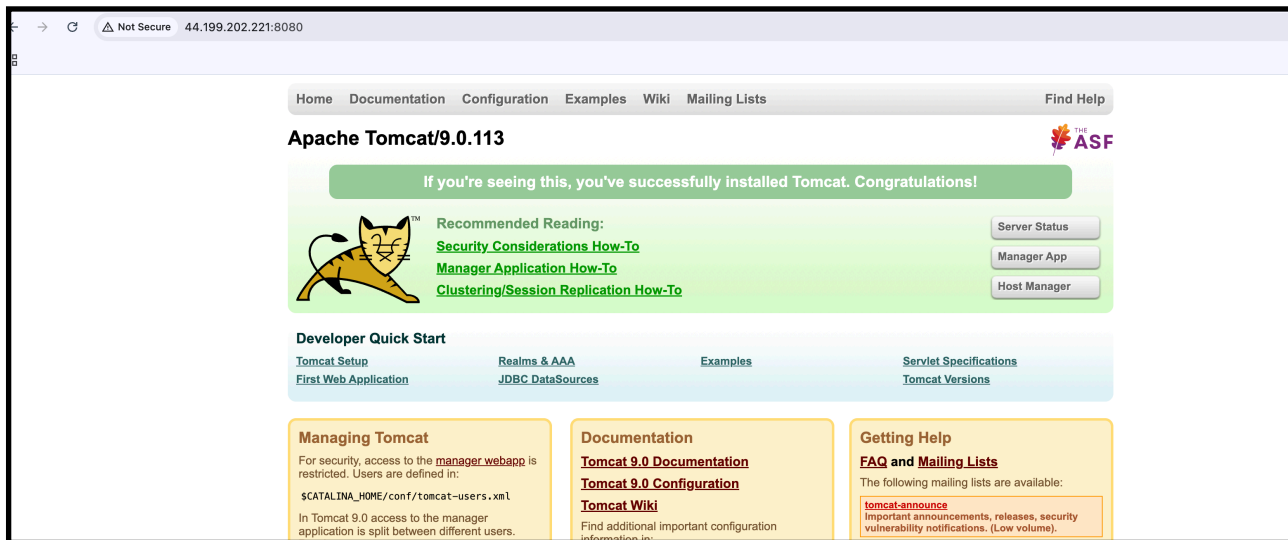
```
[root@ip-172-31-10-35 opt]# ls
apache-tomcat-9.0.113.tar.gz  aws
[root@ip-172-31-10-35 opt]# tar -xvzf apache-tomcat-9.0.113.tar.gz
apache-tomcat-9.0.113/conf/
apache-tomcat-9.0.113/conf/catalina.policy
apache-tomcat-9.0.113/conf/catalina.properties
apache-tomcat-9.0.113/conf/context.xml
```

```
[root@ip-172-31-10-35 opt]# ls
apache-tomcat-9.0.113  apache-tomcat-9.0.113.tar.gz  aws
[root@ip-172-31-10-35 opt]# mv apache-tomcat-9.0.113 tomcat
[root@ip-172-31-10-35 opt]# ls
apache-tomcat-9.0.113.tar.gz  aws  tomcat
[root@ip-172-31-10-35 opt]#
```

```
[root@ip-172-31-10-35 tomcat]# cd bin
[root@ip-172-31-10-35 bin]# bash startup
bash: startup: No such file or directory
[root@ip-172-31-10-35 bin]# bash startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-10-35 bin]#
```

--> tomcat started

Check on the web with the public ip



- Now create user in tomcat.
- Take file webapps and edit for uncomment the four files
- find / -name context.xml

```
Tomcat started.
[root@ip-172-31-10-35 bin]# find / -name context.xml
/opt/tomcat/conf/context.xml
/opt/tomcat/webapps/docs/META-INF/context.xml
/opt/tomcat/webapps/examples/META-INF/context.xml
/opt/tomcat/webapps/host-manager/META-INF/context.xml
/opt/tomcat/webapps/manager/META-INF/context.xml
[root@ip-172-31-10-35 bin]#
```

```
vi /opt/tomcat/webapps/docs/META-INF/context.xml
<!-- & -->
```

```

limitations under the License.
-->
<Context antiResourceLocking="false" >
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
</Context>
~
~
~

```

vi /opt/tomcat/webapps/examples/META-INF/context.xml

```
<!-- & -->
```

```

<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context>
    <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
        sameSiteCookies="strict" />
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,::1/128" /> -->
</Context>
~
~
~

```

vi /opt/tomcat/webapps/host-manager/META-INF/context.xml

```
<!-- & -->
```



```
<role rolename="manager-gui"/>

<role rolename="manager-script"/>

<role rolename="manager-jmx"/>

<role rolename="manager-status"/>

<user username="admin" password="admin" roles="manager-gui,manager-script,manager-
jmx,manager-status"/>

<user username="deployer" password="deployer" roles="manager-script"/>

<user username="tomcat" password="s3cret" roles="manager-gui"/>
```

```
-->
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>

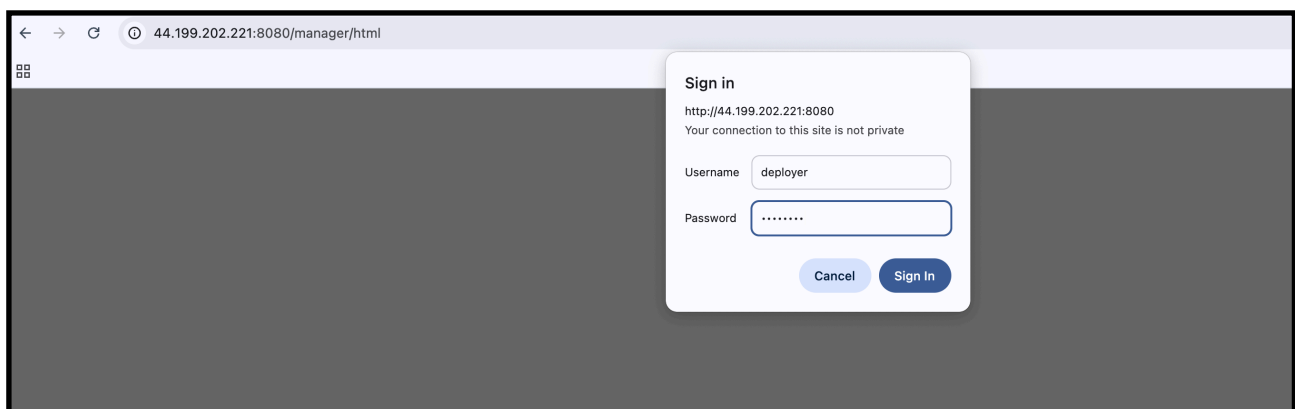
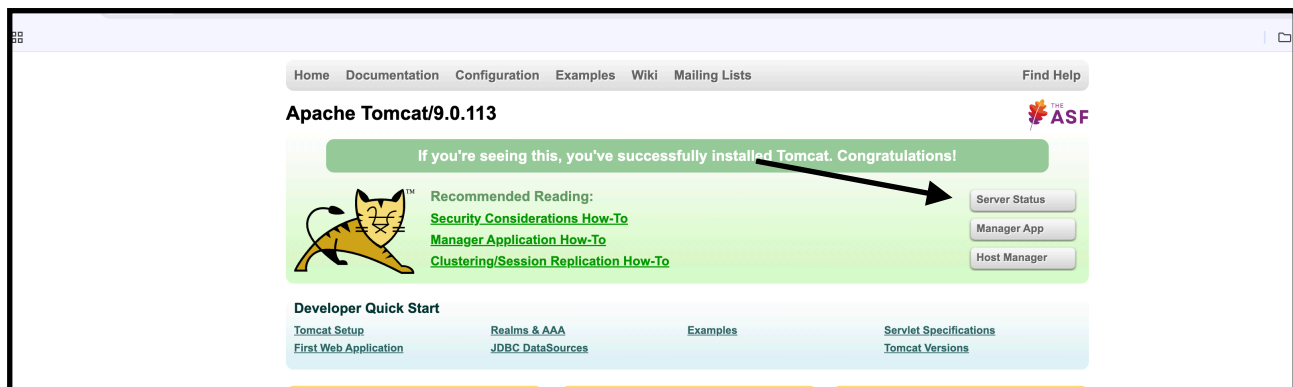
<user username="admin" password="admin" roles="manager-gui,manager-script,manager-jmx,manager-status"/>
<user username="deployer" password="deployer" roles="manager-script"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
</tomcat-users>
~
INSERT
```

After editing

Shutdown and start the tomcat

```
[root@ip-172-31-10-35 bin]# bash shutdown.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --
--add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --
[root@ip-172-31-10-35 bin]# bash startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-10-35 bin]#
```

Now in web page click on server status



Goto the tomcat page click on server status

Username – deployer

Password – deployer



Server Status

Manager

[List Applications](#)[HTML Manager Help](#)[Manager Help](#)[Complete Server Status](#)

Server Information

Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/9.0.113	17.0.17+10-LTS	Amazon.com Inc.	Linux	6.1.158-180.294.amzn2023.x86_64	amd64	ip-172-31-10-35.ec2.internal	172.31.10.35

JVM

Free Memory: 7.82 MiB Total Memory: 26.26 MiB Max Memory: 222.37 MiB

Memory Pool	Type	Initial	Total	Maximum	Used
Eden Space	Heap memory	4.31 MiB	7.31 MiB	61.37 MiB	5.20 MiB (8%)
Survivor Space	Heap memory	0.50 MiB	0.87 MiB	7.62 MiB	0.50 MiB (6%)
Tenured Gen	Heap memory	10.68 MiB	18.07 MiB	153.37 MiB	12.73 MiB (8%)
CodeHeap 'non-nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	5.55 MiB	1.21 MiB (21%)
CodeHeap 'non-profiled nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	117.22 MiB	1.61 MiB (1%)
CodeHeap 'profiled nmethods'	Non-heap memory	2.43 MiB	8.00 MiB	117.21 MiB	7.98 MiB (6%)
Compressed Class Space	Non-heap memory	0.00 MiB	2.50 MiB	1024.00 MiB	2.36 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	25.00 MiB	-0.00 MiB	24.67 MiB

"http-nio-8080"

Max threads: 200 Current thread count: 10 Current threads busy: 1 Keep alive sockets count: 3
Max processing time: 1126 ms Processing time: 1.613 s Request count: 16 Error count: 9 Bytes received: 0.00 MiB Bytes sent: 0.09 MiB

Stage	Time	Bytes Sent	Bytes Recv	Client (Forwarded)	Client (Actual)	VHost	Request
S	3 ms	0 KiB	0 KiB	14.192.14.55	14.192.14.55	44.199.202.221	GET /manager/status HTTP/1.1

?: Parse and prepare request S: Service F: Finishing R: Ready K: Keepalive

Copyright © 1999-2025, Apache Software Foundation

—> Now Just go to jenkins add these credentials deployer and take WAR/EAR files as ****/*.war**

Add the same credentials

Username – deployer , Password – deployer

 Jenkins

hiring-app

Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Filter

Aggregate downstream test results

Archive the artifacts

Build other projects

Publish JUnit test result report

Record fingerprints of files to track usage

Git Publisher

Deploy war/ear to a container

E-mail Notification

Editable Email Notification

Set GitHub commit status (universal)

Set build status on GitHub commit [deprecated]

Slack Notifications

Delete workspace when build is done

+ Add post-build action

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Deploy war/ear to a container

WAR/EAR files ?

**/*.war

Context path ?

Containers

Tomcat 9.x Remote

Credentials

deployer/***** (tomcat_credentials)

+ Add

Tomcat URL ?

http://44.199.202.221:8080/

Advanced

+ Add Container

☐ Deploy on failure

Save

Apply

Configure

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Jenkins Credentials Provider: Jenkins

Domain

Global credentials (unrestricted)

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

deployer

☐ Treat username as secret ?

Password ?

.....

ID ?

tomcat_credentials

Description ?

tomcat_credentials

Slack Notifications

—> and save

—> After save Run Build

Jenkins / hiring-app

Status

Changes

Workspace

Build Now

Configure

Delete Project

SonarQube

Rename

Builds

Filter

Today

#16 11:43 AM

hiring-app

SonarQube

SonarQube Quality Gate

Sabear **Passed**

server-side processing: **Success**

Permalinks

- Last build (#15), 1 hr 46 min ago
- Last stable build (#15), 1 hr 46 min ago
- Last successful build (#15), 1 hr 46 min ago
- Last failed build (#14), 1 hr 50 min ago
- Last unsuccessful build (#14), 1 hr 50 min ago
- Last completed build (#15), 1 hr 46 min ago

```

GroupID: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:hiring-app
Uploading: http://44.211.69.235:8081/repository/hiring-app/in/javahome/hiring /0.1/hiring -0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://44.211.69.235:8081/repository/hiring-app/in/javahome/hiring /0.1/hiring -0.1.war (1.7 kB at 24 kB/s)
Uploading artifact hiring.war completed.
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/hiring-app/target/hiring.war to container Tomcat 9.x Remote with context null
    [/var/lib/jenkins/workspace/hiring-app/target/hiring.war] is not deployed. Doing a fresh deployment.
    Deploying [/var/lib/jenkins/workspace/hiring-app/target/hiring.war]
[Slack Notifications] found #15 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS

```

- Now our deployment successfully done
- Now you can check in tomcat cli webapps one file is shown
- Cd /opt/tomcat/webapps/hiring

```

tomcat started.
[root@ip-172-31-10-35 bin]# cd /opt/tomcat/webapps/hiring
[root@ip-172-31-10-35 hiring]# ls
META-INF  WEB-INF  index.jsp
[root@ip-172-31-10-35 hiring]# cd ..
[root@ip-172-31-10-35 webapps]# ll
total 36
drwxr-x---.  3 root root 16384 Jan  2 10:26 ROOT
drwxr-x---. 16 root root 16384 Jan  2 10:26 docs
drwxr-x---.  7 root root   99 Jan  2 10:26 examples
drwxr-x---.  4 root root   54 Jan  2 11:43 hiring
-rw-r-----.  1 root root  1698 Jan  2 11:43 hiring.war
drwxr-x---.  6 root root   79 Jan  2 10:26 host-manager
drwxr-x---.  6 root root  114 Jan  2 10:26 manager
[root@ip-172-31-10-35 webapps]#

```

And you can check in tomcat at browser we got our application is live.

